RESOLUTION MSC.36(63)
(adopted on 20 May 1994)
ADOPTION OF THE INTERNATIONAL CODE OF SAFETY
FOR HIGH SPEED CRAFT MSC 63/23/Add.2
ANNEX 18
Page 206
HL1 Heeling
Multihull craft criteria
At
RLl =0 HeaUAq lever d"a tD vind ,
ILT'- # HealiAq lever due to villCi ,CJ1ISUJl9

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HL1 Heeling
Multihull craft criteria
At
RLl =0 HeaUAq lever d"a tD vind ,
ILT'- # HealiAq lever due to villCi ,CJ1ISUJl9
 (passenge~ crowding or turnlng)
Figure 1 - Intact stability
Figure 2 - Damage stability
lever due to wind
HTL = Heeling lever due to wind + gusting + (passenger crowding or
turning)
HL3 Heeling lever due to wind
HL4 = Heeling lever due to wind + passenger crowding
em = Angle of maximum GZ
ed Angle of downflooding
er Angle of roll
ee = Angle of equilibrium, assuming no wind, passenger crowding or
turning effects
eh = Angle of heel due to heeling lever HL1, HTL, HL3 or HL4
Al ~ Area required by 1.1
A2 ~ 0.028 m.rad
W/1689n/pp/EWP
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ADOPTION OF THE INTERNATIONAL CODE OF SAFETY
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DEFINITIONS, REQUIREMENTS AND COMPLIANCE CRITERIA
RELATED TO OPERATIONAL AND SAFETY PERFORMANCE
This annex applies to all types of craft. Tests to evaluate operational
safety should be conducted on the prototype craft of a new design or of a

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DEFINITIONS, REQUIREMENTS AND COMPLIANCE CRITERIA
RELATED TO OPERATIONAL AND SAFETY PERFORMANCE
This annex applies to all types of craft. Tests to evaluate operational safety should be conducted on the prototype craft of a new design or of a design incorporating new features which may modify the results of a previous testing. The tests should be carried out to a schedule agreed between the Administration and the manufacturer. Where conditions of service warrant additional testing (e.g., low temperature), the Administration or base port State authorities as appropriate may require further demonstrations.
Functional descriptions, technical and system specifications relevant to the understanding and evaluation of craft performance should be available.
The objective of these tests is to provide essential information and guidance to enable the craft to be operated safely under normal and emergency conditions within the design speed and environmental envelope.
The following procedures are outlined as requirements in dealing with verification of craft performance.
1 Performance
1.1 General
1.1.1 The craft should meet the applicable operational requirements in

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The following procedures are outlined as requirements in dealing with verification of craft performance.
1 Performance
1.1 General
1.1.1 The craft should meet the applicable operational requirements in chapter 17 of this Code and this annex for all extremes of passenger and load configurations for which certification is required. The limiting sea state related to the different modes of operation should be verified by tests and analyses of a craft of the type for which certification is requested.
1.1.2 Operational control of the craft should be in accordance with procedures established by the applicant for operation in service. Procedures to be established should be start procedure, cruise procedures, normal and emergency stop and manoeuvre procedures.
1.1.3 The procedures established under 1.1.2 should:
1.1.4
1.1.5
.1 demonstrate that normal manoeuvres and craft responses to failures are consistent in performance;
.2 use methods or devices that are safe and reliable; and
.3 include allowance for any time lag in the execution of procedures that may reasonably be expected in service.
Procedures required by this annex should be conducted over water of

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.2 use methods or devices that are safe and reliable; and
.3 include allowance for any time lag in the execution of

procedures that may reasonably be expected in service.
Procedures required by this annex should be conducted over water of sufficient depth such that craft performance will not be affected.
Tests should be conducted at minimum practicable weight and additional testing should be conducted at maximum weight sufficient to establish the need for additional restrictions and for testing to examine the effect of weight.
W/1689n/pp/EWP
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2 Stopping
..
2.1 This test is to establish the acceleration experienced when stopping the craft in calm water with no passenger load or cargo load during the following conditions:
.1 normal stop for maximum operational speed;
.2 emergency stop for maximum operational speed; and
.3 crash stop from maximum operational speed and from any transient mode speed.
2.2 The tests referred to in 2.1.1 and 2.1.2 should document that the

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SOURCE: MSC.36(63).pdf
.2 emergency stop for maximum operational speed; and
.3 crash stop from maximum operational speed and from any transient mode speed.
2.2 The tests referred to in 2.1.1 and 2.1.2 should document that the
accelerations do not exceed safety level 1 in annex 3 when control levers are used in accordance to written procedures as given in the craft operating manual or in an automatic mode. Should safety level 1 be exceeded during normal stop, control systems should be modified in order to avoid exceedance or passengers should be required to be seated during normal stop. Should safety level 1 be exceeded during emergency stop, then written procedures in the craft operating manual should include detailed information of how to avoid exceedance or control system should be modified to avoid exceedance.
2.3 The test referred to in 2.1.3 should document that the accelerations do not exceed safety level 2 of annex 3 when control levers of automatic modes are used in a manner which will give the highest accelerations. If safety level 2 is exceeded then the craft operating manual should include a warning that it is a risk to passengers being injured, if crash stop is performed.

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modes are used in a manner which will give the highest accelerations. If safety level 2 is exceeded then the craft operating manual should include a warning that it is a risk to passengers being injured, if crash stop is

performed.
2.4 Other tests should be repeated during craft turning to establish the need or otherwise to impose any speed related restrictions during manoeuvres.
3 ~se performance
3.1 This test is to establish the craft performance and accelerations experienced during cruise modes with no passenger load or cargo load during the following conditions:
.1 normal operation conditions are those in which the craft will safely cruise at any heading while manually operated, auto-pilot assisted operated or operated with any automatic control system in normal mode; and
.2 worst intended conditions, referred to in 1.4.48 of this Code, are those in which it should be possible to maintain safe cruise without exceptional piloting skill. However, operations at all headings relative to the wind and sea may not be possible. For type of craft having a higher performance standard in non-displacement mode, the

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exceptional piloting skill. However, operations at all headings relative to the wind and sea may not be possible. For type of craft having a higher performance standard in non-displacement mode, the